Quick Phone Repair Management System

Requirements Catalogue and Memo | Aligned with Sprint 2 and Sprint 3 presentations | MIS 4173 | October 23, 2026

Revision memo

To: Quick Phone project stakeholders and MIS 4173 instructor

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Subject: Revised requirements, AI traceability, and Sprint 3 design alignment

We updated our Sprint 2 requirements after reviewing the instructor feedback and both presentations. The main catalogue now has one list of requirements, a source column, and a separate acceptance check for each item. We kept the original FR01–FR14 numbering because the Sprint 3 interface slides refer to those IDs. New client-confirmed rules are numbered FR15–FR21, so the prototype references do not change.

The client-confirmed decisions remain the basis for design: four staff roles; repair tracking from intake through pickup; automatic inventory deduction when technicians record parts used; Owner approval for every purchase order; and completion-only email using the store account. We also kept appointment scheduling for planned drop-offs while allowing walk-ins to skip an appointment, as shown in Sprint 3.

Appendix A maps the ten important findings to the final catalogue, and Appendix B preserves the original counts. The original 31-row AI output was not included with the earlier submission, so we do not present a reconstructed list as the original.

Sprint 3 moves from logical work to implementation choices. Its presentation describes seven DFD data stores and a simplified 16-entity physical ERD. Those are design decisions, not 16 new requirements. The existing requirement IDs are used to trace the three interface designs and the physical models. Some details still need client confirmation; those are identified rather than silently treated as approved.

Source key and priorities

CC = client-confirmed rule; INT = interview notes; PM = Sprint 1 process model; S2 = Sprint 2 use cases/presentation; S3 = Sprint 3 design/presentation; AI-added = missing rule highlighted through AI review; AI-refined = existing rule clarified after review; AI-highlighted = already present but emphasized by AI. “Must” supports the agreed workflow; “Should” is useful but can be sequenced after core functions. A proposed design detail is not automatically a client-approved requirement.

Consolidated functional requirements (FR01–FR21)

ID	Area	Priority	Requirement	Source	Acceptance check
FR01	Appointments	Must	Front Desk can schedule or change a device drop-off appointment and assign a technician. Walk-in repairs can be checked in without an appointment.	INT / PM / S2 / S3	Schedule and reschedule one appointment, then create a walk-in repair without an appointment. Check both records.
FR02	Device check-in	Must	Front Desk creates a repair job linked to a customer and device, and records the issue and intake condition.	INT / PM / S2 / S3	Create a repair job; reopen it and confirm the customer, device, problem and condition are correct.
FR03	Claim ticket and tag	Should	The system creates a claim ticket and device tag with the same repair job number.	PM / S2 / S3	Print a ticket and tag for a sample job and confirm both match its job number.
FR04	Technician assignment	Must	Front Desk can assign a repair to a Technician; Technicians can see the work assigned to them.	PM / S2 / S3 / AI-highlighted	Assign a job and confirm it appears only in the correct technician queue.
FR05	Diagnosis and quote	Must	A Technician can record a diagnosis and quote, including labor and parts. Staff record the customer quote decision before repair continues.	PM / S2 / S3 / AI-refined	Save a diagnosis and quote, record an approval/rejection, and confirm the correct next status.
FR06	Repair status	Must	Authorized staff can track a job from check-in through diagnosis, approval, repair, ready for pickup and release/closure, keeping a history of changes.	CC / S2 / S3 / AI-refined	Move a sample repair through its main statuses; verify current status and history.
FR07	Parts during repair	Must	A Technician can check parts available at the store and record actual parts used. Recording use automatically deducts on-hand stock; a negative balance must be prevented.	CC / PM / S2 / S3 / AI-refined	Record use of two parts and verify two are deducted. Try using more than available and confirm the system blocks it.
FR08	Pickup and payment	Must	Front Desk records payment status, verifies the device recipient and closes the repair when it is released.	PM / S2 / S3 / AI-highlighted	Record pickup and payment status on a finished job, verify recipient check and closed status.
FR09	Inventory records	Must	Manager can review stock by store and record receipts, usage and corrections; inventory quantities stay consistent with logged transactions.	PM / S2 / S3	Post a receipt and adjustment and reconcile on-hand stock against transaction history.
FR10	Purchasing	Must	Manager can draft purchase orders for needed parts and suppliers. Every purchase order needs Owner approval before it proceeds.	CC / PM / S2 / S3 / AI-added	Create a PO as Manager, submit it and verify that it cannot proceed without Owner approval.
FR11	CSV part import	Should	Manager can import a CSV of part records and see validation errors for invalid rows.	PM / S2 / S3	Import valid rows; test an invalid row and confirm an understandable error.
FR12	Customer alerts	Must	The system sends one completion-only email notification from the store email account when repair is completed. Technicians cannot send customer email; Owner, Manager and Front Desk have notification access.	CC / S2 / S3 / AI-refined	Complete a repair and verify one store-sent completion email. Verify no email for earlier statuses and test Technician restrictions.
FR13	Owner dashboard	Must	Owner can view open repairs, repair status and inventory information across the stores.	PM / S2 / S3	Compare dashboard repair counts and stock figures to test records from both stores.
FR14	Users and stores	Should	Owner or Manager can maintain staff accounts with one of four approved roles and a store assignment where applicable; permissions are role-based.	CC / PM / S2 / S3 / AI-refined	Create accounts for Owner, Manager, Technician and Front Desk; verify roles and store access.
FR15	Owner-only PO decision	Must	Only the Owner can approve or reject a submitted purchase order, and the decision is recorded.	CC / S2 / S3 / AI-added	Try approval as Manager and as Owner; only the Owner action should work and the decision is saved.
FR16	Store sender account	Must	Customer repair-completion messages use Quick Phone's configured store email address as the sender.	CC / S2 / S3 / AI-added	Send a completion email and verify the sender address matches the store configuration.
FR17	Notification access	Must	Owner, Manager and Front Desk can access permitted notification actions; Technicians cannot send customer notifications or view unnecessary customer contact details.	CC / S2 / S3 / AI-added	Test notification actions for all four roles and confirm the Technician is blocked.
FR18	Low-stock warnings	Should	Manager can identify parts at or below their reorder point for each store.	S2 / S3	Set test reorder points and verify the correct low-stock items appear.
FR19	Waiting for parts	Should	When a needed part is out of stock, the job can be marked Awaiting Parts and linked to a draft order; receiving stock allows staff to resume work.	S3 design assumption	Create an out-of-stock case and verify the wait status, linked PO and restart after receipt.
FR20	Receiving POs	Should	Authorized staff can record full or partial receipt of an approved purchase order and update inventory accordingly.	S3 design assumption	Record a partial delivery against an approved PO; confirm received quantities and stock change.
FR21	Customer reuse	Should	Front Desk can search for an existing customer before creating a new record to reduce duplicate entries.	S3 design assumption	Look up a customer by phone number and reuse the record when registering a repair.

Nonfunctional requirements

ID	Area	Priority	Requirement	Source	Acceptance check
NFR01	Authorization	Must	Role permissions prevent unauthorized access to email actions, purchasing approvals and protected records.	CC / S2 / S3	Sign in under all four roles and test restricted actions.
NFR02	Data consistency	Must	Repair status, inventory quantities, purchasing and payment records remain consistent after updates.	PM / S2 / S3	Perform a repair, part deduction and PO receipt; compare saved records and stock totals.
NFR03	Usability	Should	Front Desk and Technician interfaces provide readable forms and queues for common tasks.	PM / S2 / S3	Have a user complete check-in and diagnosis from the prototypes; record confusing steps.
NFR04	Privacy	Must	Customer contact and device details are shown only to staff with a work-related need.	INT / S2 / S3	Check which customer fields each role sees, including Technician access.

Sprint 2 / Sprint 3 presentation crosswalk

Slide/topic	Aligned requirements	Connection / check
Sprint 2: scheduling, intake and release	FR01–FR08, FR12; NFR01–04	Logical DFD: Check In, Diagnose, Repair, Notify Customer, Release; use cases UC-01–UC-06
Sprint 2: stock, purchasing, owner view	FR07, FR09–FR10, FR13–FR18	Logical DFD: Replenish Parts, Report Repair Status, Maintain Shop Records; UC-07–UC-11
Sprint 3: Front Desk interface, slide 7	FR01–FR04, FR21, NFR03	Prototype cites FR01–FR04. Search-before-create is extra S3 design detail; check-in process naming varies between slides.
Sprint 3: Technician interface, slide 8	FR04–FR07, FR17, FR19, NFR01, NFR02, NFR04	Prototype cites FR04–FR07. Technician privacy and Awaiting Parts are additionally represented.
Sprint 3: Manager interface, slide 9	FR09–FR11, FR15, FR18–FR20, NFR01–02	Prototype cites FR09–FR11. Includes PO approval, reorder threshold, partial receipts and waiting jobs.
Sprint 3: physical DFD / ERD, slides 3–6	FR02, FR05–FR10, FR12–FR17, NFR02	Presentation describes 7 data stores holding 16 physical ERD entities. Verify exact table mapping against final models.

Appendix A — Ten salient AI findings mapped into the catalogue

#	AI finding	Current requirement ID	Treatment
1	Technician assignment and assigned queue	FR04	AI-highlighted
2	Quote and customer approval	FR05	AI-refined
3	Intake-to-closure tracking	FR06	AI-refined
4	Automatic deduction on actual use	FR07	AI-refined
5	Payment, pickup and release	FR08	AI-highlighted
6	Owner PO approval	FR10 / FR15	AI-added
7	Completion-only email	FR12	AI-refined
8	Notification role restrictions	FR12 / FR17	AI-added
9	Store email as sender	FR12 / FR16	AI-added
10	Four approved staff roles	FR14	AI-refined

Appendix B — Original AI comparison counts

Disposition in earlier memo	Count	Interpretation
Salient	10	Mapped to specific requirement IDs in Appendix A.
Overlapping	11	Existing requirements were not duplicated.

Unsupported / spurious	7	Not accepted as confirmed rules without supporting evidence.
Conflicting	3	Resolved in favor of client-confirmed rules.
Total	31	Original 31-item AI list not provided; these counts are from the earlier memo.